

Effect of a Quality Improvement Intervention on Clinical Outcomes in Patients in India With Acute Myocardial Infarction

The ACS QUIK Randomized Clinical Trial

Mark D. Huffman, MD, MPH; Padinhare P. Mohanan, MD, DM; Raji Devarajan, MS; Abigail S. Baldrige, MS; Dimple Kondal, PhD; Lihui Zhao, PhD; Mumtaj Ali, MCA; Mangalath N. Krishnan, MD, DM; Syam Natesan, MBBS, MD, DM; Rajesh Gopinath, MD, DM; Sunitha Viswanathan, MD, DM; Joseph Stigi, MD, DM; Johnny Joseph, MD, DM; Somanathan Chozhakkat, MD, DM; Donald M. Lloyd-Jones, MD, ScM; Dorairaj Prabhakaran, MD, DM, MSc; for the Acute Coronary Syndrome Quality Improvement in Kerala (ACS QUIK) Investigators

IMPORTANCE Wide heterogeneity exists in acute myocardial infarction treatment and outcomes in India.

OBJECTIVE To evaluate the effect of a locally adapted quality improvement tool kit on clinical outcomes and process measures in Kerala, a southern Indian state.

DESIGN, SETTING, AND PARTICIPANTS Cluster randomized, stepped-wedge clinical trial conducted between November 10, 2014, and November 9, 2016, in 63 hospitals in Kerala, India, with a last date of follow-up of December 31, 2016. During 5 predefined steps over the study period, hospitals were randomly selected to move in a 1-way crossover from the control group to the intervention group. Consecutively presenting patients with acute myocardial infarction were offered participation.

INTERVENTIONS Hospitals provided either usual care (control group; n = 10 066 participants [step 0: n = 2915; step 1: n = 2649; step 2: n = 2251; step 3: n = 1422; step 4: n = 829; step 5: n = 0]) or care using a quality improvement tool kit (intervention group; n = 11 308 participants [step 0: n = 0; step 1: n = 662; step 2: n = 1265; step 3: n = 2432; step 4: n = 3214; step 5: n = 3735]) that consisted of audit and feedback, checklists, patient education materials, and linkage to emergency cardiovascular care and quality improvement training.

MAIN OUTCOMES AND MEASURES The primary outcome was the composite of all-cause death, reinfarction, stroke, or major bleeding using standardized definitions at 30 days. Secondary outcomes included the primary outcome's individual components, 30-day cardiovascular death, medication use, and tobacco cessation counseling. Mixed-effects logistic regression models were used to account for clustering and temporal trends.

RESULTS Among 21 374 eligible randomized participants (mean age, 60.6 [SD, 12.0] years; n = 16 183 men [76%]; n = 13 689 [64%] with ST-segment elevation myocardial infarction), 21 079 (99%) completed the trial. The primary composite outcome was observed in 5.3% of the intervention participants and 6.4% of the control participants. The observed difference in 30-day major adverse cardiovascular event rates between the groups was not statistically significant after adjustment (adjusted risk difference, −0.09% [95% CI, −1.32% to 1.14%]; adjusted odds ratio, 0.98 [95% CI, 0.80-1.21]). The intervention group had a higher rate of medication use including reperfusion but no effect on tobacco cessation counseling. There were no unexpected adverse events reported.

CONCLUSIONS AND RELEVANCE Among patients with acute myocardial infarction in Kerala, India, use of a quality improvement intervention compared with usual care did not decrease a composite of 30-day major adverse cardiovascular events. Further research is needed to understand the lack of efficacy.

TRIAL REGISTRATION clinicaltrials.gov Identifier: [NCT02256657](https://clinicaltrials.gov/ct2/show/study/NCT02256657)

JAMA. 2018;319(6):567-578. doi:10.1001/jama.2017.21906

◀ Editorial [page 554](#)

◀ Related article [page 607](#)

+ Supplemental content

Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: The Acute Coronary Syndrome Quality Improvement in Kerala (ACS QUIK) Investigators are listed at the end of this article.

Corresponding Author: Mark D. Huffman, MD, MPH, Northwestern University Feinberg School of Medicine, 680 N Lake Shore Dr, Ste 1400, Chicago, IL 60611 (m-huffman@northwestern.edu).

In 2015, there were an estimated 7.3 million (95% uncertainty interval, 6.8 million–7.8 million) fatal myocardial infarctions globally, and South Asia was estimated to have the world's highest age-standardized incident rate of myocardial infarction.¹ High-income countries have developed programs for improving process and outcomes measures for acute myocardial infarction that have been associated with improvements in care and outcomes with concomitant reductions in racial/ethnic disparities in care.² Because ischemic heart disease represents the leading cause of global deaths,¹ improving the quality of care and outcomes for patients with acute myocardial infarction, particularly in low- and middle-income countries, is a global health priority.³

Previous publications on the presentation, management, and outcomes of 25 748 acute coronary syndrome patients from 125 hospitals in Kerala, India,⁴ have demonstrated wide heterogeneity in process and outcome measures across hospitals⁵ and identified targets for intervention, including increasing speed and use of guideline-directed medical therapy. Previous acute myocardial infarction quality improvement randomized trials that include physician education, audit and feedback mechanisms,⁶ clinical pathways,⁷ and checklists⁸ have shown favorable effects on process measures in low- and middle-income countries such as Brazil⁹ and China.¹⁰ However, these trials were not designed or powered to evaluate the effects of these interventions on clinical outcomes, nor were they designed to account for temporal trends. A 2017 pre/post-intervention study in Tamil Nadu, India, also demonstrated favorable trends with implementation of a hub-and-spoke health system intervention but used a nonrandomized study design.¹¹

To determine whether process and outcome measures could be improved for patients with acute myocardial infarction with a locally adapted quality improvement tool kit using a robust study design for better causal inference, we performed a large, pragmatic, cluster randomized, stepped-wedge trial among 63 hospitals in Kerala.

Methods

Study Design

The Acute Coronary Syndrome Quality Improvement in Kerala (ACS QUIK) trial was a pragmatic, cluster randomized, stepped-wedge clinical trial in which hospitals were randomized to receive the quality improvement tool kit intervention at 1 of 5 predefined, 4-month steps over a 24-month period between November 10, 2014, and November 9, 2016, after a period of usual care. The last date of follow-up was December 31, 2016. The methods have been previously published.¹² A stepped-wedge design is useful when evaluating an intervention when clusters have ethical concerns about being randomized to the control group only for the duration of the trial and when it is infeasible to disseminate the intervention simultaneously across a large number of clusters. The study received ethics board approval from local, national, and international bodies and was approved by the Indian Health Ministry Screening Committee. All participants or their prox-

Key Points

Question What is the effect of a locally adapted quality improvement tool kit for acute coronary syndrome on major adverse cardiovascular events in Kerala, India?

Findings In this cluster randomized, stepped-wedge clinical trial that included 63 hospitals and 21 374 participants with acute myocardial infarction, the 30-day composite event rates were 6.4% in the control phase and 5.3% in the intervention phase, a difference that did not reach statistical significance after adjustment for cluster and temporal trends.

Meaning Among patients with acute myocardial infarction in Kerala, India, a locally adapted quality improvement tool kit did not reduce the rate of 30-day major adverse cardiovascular events compared with usual care.

ies provided written informed consent to participate. The study was conducted according to the study protocol ([Supplement 1](#)) and analyzed according to the statistical analysis plan ([Supplement 2](#)).

Hospitals and Study Participants

We recruited 63 hospitals in Kerala from a sample ($n = 125$) that had previously participated in the Kerala Acute Coronary Syndrome (ACS) Registry.⁴ We also sought government hospitals that had not participated in this registry to include a range of hospital types (ie, government, private, nonprofit/charity). To be eligible, hospitals had to identify 2 individuals who would be willing to serve on a quality improvement team. Patients were eligible to participate if they presented or were transferred for evaluation and management of either non-ST-segment elevation myocardial infarction (NSTEMI) or ST-segment elevation myocardial infarction (STEMI) based on the Third Universal Definition of Myocardial Infarction.¹³

Randomization and Treatment Assignments

The study biostatisticians (D.K. and L.Z.) performed central computer-based randomization of hospitals. The other members of the study team and the selected sites were informed of the 12 or 13 sites that would cross over to the intervention period 2 weeks before each of the predefined steps to maintain allocation concealment while aiding in training logistics. We stratified randomization into 4 groups by projected recruitment (≤ 200 , 201–500, 501–1000, and >1000 participants) based on the Kerala ACS Registry⁴ to minimize potential imbalance between the intervention and control periods. By nature of the trial design, neither personnel nor participants were blinded to the intervention.

Interventions

We used formative mixed-methods research and evidence synthesis to adapt previously reported strategies with the goal of improving process of care and outcome measures. We created an audit and feedback reporting mechanism based on key data elements used by the American College of Cardiology and American Heart Association.¹⁴ These reports,

which included site-specific measures on performance and which were sent monthly via email to site investigators, compared hospital-level performance with other hospitals in each cohort and with all hospitals in the trial (see eAppendix in Supplement 3 for sample report). We trained sites on the interpretation of these reports and on the use of these reports for informing quality improvement team meetings, for which we provided meeting templates based on the Plan-Do-Study-Act cycle of change. The tool kit also included standardized admission and discharge order sets; translated and culturally adapted patient education materials related to tobacco cessation, dietary advice, and physical activity; and linkage to emergency cardiovascular care training based on the low prevalence of code teams among the participating hospitals (see eAppendix in Supplement 3 for sample checklists and patient education materials). We sought and received feedback from the participating investigators in the development of this tool kit and provided free online health care quality and patient safety training to sites and their teams through the Institute for Healthcare Improvement. The control condition was usual care according to local hospital practice. We performed a concurrent mixed-methods process evaluation for assessing context, implementation fidelity, and mechanisms of impact based on the Medical Research Council's recommendations¹⁵; these results will be reported separately.

The study team performed central and on-site training for 90 to 120 minutes at each site including with the site investigator and other members of the quality improvement team and medical staff. We did not include a transition period. Date of admission was used to define allocation of participants to intervention or control.

Outcomes

The primary outcome was the composite end point of 30-day major adverse cardiovascular events, defined as death, reinfarction (defined by the Third Universal Definition of Myocardial Infarction¹³), stroke, and major bleeding (defined by the Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries [GUSTO] criteria,¹⁶ which is defined by intracerebral hemorrhage or bleeding resulting in substantial hemodynamic compromise requiring treatment). Outcome data were collected at each site either through in-person visits or by telephone and were reported centrally. If a site was unable to reach a participant after 3 attempts, then that participant was considered lost to follow-up. Although the outcome assessors were not blinded, there did not appear to be a high risk of ascertainment bias to influence the treatment effect for objective outcomes such as these.¹⁷

Secondary outcomes included 30-day all-cause mortality, 30-day cardiovascular mortality, in-hospital mortality, 30-day myocardial reinfarction, 30-day stroke, 30-day major GUSTO bleeding, optimal in-hospital medication use (composed of aspirin, adenosine diphosphate receptor antagonist [clopidogrel, prasugrel, or ticagrelor], anticoagulant, and β -blocker; in-hospital statin use was additionally predefined but data were not collected), optimal discharge

medication use (composed of aspirin, adenosine diphosphate receptor antagonist [clopidogrel, prasugrel, or ticagrelor], statin, and β -blocker), and tobacco cessation advice. We performed post hoc analyses evaluating the effect of the intervention on major adverse cardiovascular events plus in-hospital incident heart failure, cardiogenic shock, and cardiac arrest.

Statistical Analysis

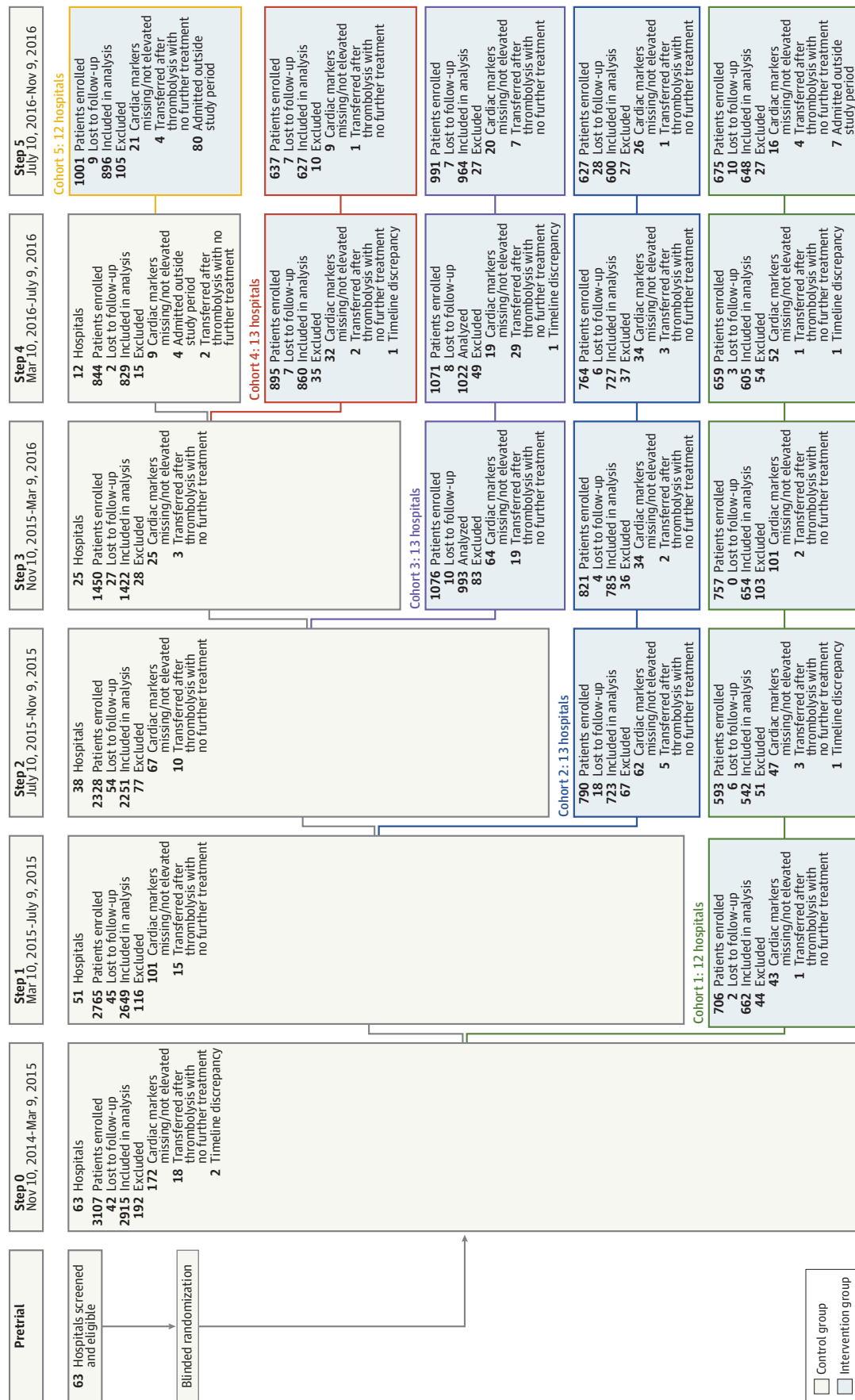
We initiated the trial with a target sample size of 15 750 participants from 63 hospitals with 5 steps to have 80% power to achieve a 2.4% absolute reduction in major adverse cardiovascular events with a 2-sided $\alpha = .05$.¹⁸ We assumed a rate of loss to follow-up of 5% and an intraclass correlation coefficient of 0.05 based on data from the Kerala ACS Registry (unpublished data) and a previous quality improvement trial in Brazil.¹⁹ This intervention effect estimate was based on the difference between leading and lagging hospitals' performance and outcomes from previous observational research.²⁰ Prespecified participant-level results are reported.

Baseline characteristics (unadjusted and adjusted for within-hospital clustering and temporal trends) were summarized for intervention and control groups. For the primary outcome, the overall difference in 30-day major adverse cardiovascular event rates between control and intervention periods was reported. In the primary analysis, 30-day major adverse cardiovascular events were modeled using mixed-effects logistic regression with a random cluster (hospital) effect and a fixed time effect for every 4-month step. As a secondary analysis, the model was also adjusted for age, sex, type of myocardial infarction, systolic blood pressure, heart rate, serum creatinine level, acute heart failure, cardiogenic shock, and resuscitated cardiac arrest, which are covariates included in the Global Registry of Acute Coronary Events (GRACE) risk score.²¹ We performed post hoc analyses to evaluate the potential effect of the intervention on an expanded definition of major adverse cardiovascular events to include cardiogenic shock, incident heart failure, and cardiac arrest and the interaction between time spent in the intervention period and the adjusted 30-day major adverse cardiovascular event rate. Results are also reported by prespecified subgroups of participant age, sex, STEMI vs NSTEMI, hospital size, and hospital type, for which we also tested interaction effects.

All results are reported using an intention-to-treat analysis. The interim analysis was performed at 12 months for reporting to the data and safety monitoring board, but only the study biostatisticians (D.K. and L.Z.) were unblinded to the results. To adjust for the interim analysis, the O'Brien-Fleming stopping boundary was set at $z > 2.797$ with a 2-sided significance level of .005 for the interim analysis and $z > 1.977$ with a 2-sided significance level of .048 for the final analysis. We did not make additional a priori adjustments to the significance threshold for secondary outcomes based on multiple testing; therefore, these analyses should be considered exploratory.

Biweekly central statistical monitoring was done using previously published algorithms²² for risk-based site

Figure 1. Flow of Hospitals and Participants Through the ACS QIJK Trial



All hospitals approached agreed to participate and were included. Randomization occurred at a single time point prior to patient enrollment. Hospitals were blinded to their assigned cohort until 2 weeks prior to the time point at which their cohort moved from the control stage to intervention stage. Participants lost to follow-up include those who could not be reached after 3 contact attempts and those for whom the site did not perform the follow-up procedure. Patients lost to follow-up were included in the analysis.

Table 1. Baseline Characteristics of ACS QUIK Patients and Hospitals by Intervention and Control Group

Characteristics	Control (n = 10 066) ^a	Intervention (n = 11 308) ^a	Difference (95% CI) ^b
Patient Characteristics			
Age, mean (SD), y	60.3 (12.0)	60.9 (12.1)	0.6 (0.3 to 1.0)
Male, No. (%)	7654 (76.0)	8529 (75.4)	-0.6 (-1.8 to 0.5)
History of tobacco use, No. (%)	3772 (37.5)	2842 (25.1)	-12.3 (-13.6 to -11.1)
History of diabetes, No. (%)	4151 (41.2)	5333 (47.2)	5.9 (4.6 to 7.3)
Transferred from another facility, No. (%)	4202 (41.7)	4199 (37.1)	-4.6 (-5.9 to -3.3)
No insurance, No. (%)	6878 (68.3)	8664 (76.6)	8.3 (7.1 to 9.5)
ST-elevation myocardial infarction, No. (%)	6921 (68.8)	6768 (59.9)	-8.9 (-10.2 to -7.6)
Symptom-to-door time, median (IQR), min	240 (120-840) (n = 9711)	255 (111-825) (n = 10 849)	15 (4 to 26)
Body weight, mean (SD), kg	63.6 (9.8) (n = 10 065)	63.3 (9.6) (n = 11 304)	-0.3 (-0.5 to 0.0)
Systolic blood pressure, mean (SD), mm Hg	138.1 (29.1) (n = 10 061)	139.0 (28.9) (n = 11 303)	0.9 (0.1 to 1.7)
Heart rate, mean (SD), /min	80.0 (18.7) (n = 10 065)	79.9 (19.1) (n = 11 303)	-0.1 (-0.6 to 0.5)
Initial troponin, median (IQR), ng/mL	1.6 (0.4-8.0) (n = 4566)	1.1 (0.2-4.4) (n = 4483)	-0.5 (-0.7 to -0.4)
Low-density lipoprotein cholesterol, mean (SD), mg/dL	126.2 (40.3) (n = 7288)	119.1 (41.2) (n = 7542)	-7.1 (-8.4 to -5.8)
Triglycerides, median (IQR), mg/dL	121 (89-167) (n = 7308)	122 (90-163) (n = 7552)	1 (-1 to 3)
Serum creatinine, median (IQR), mg/dL	1.0 (0.8-1.2) (n = 4246)	1.0 (0.9-1.2) (n = 9589)	0.0 (0.0 to 0.0)
Fasting glucose, median (IQR), mg/dL	124 (98-172) (n = 6658)	130 (105-180) (n = 6740)	6 (4 to 8)
Hemoglobin, mean (SD), mg/dL	13.3 (2.0) (n = 9763)	13.2 (2.0) (n = 11 079)	-0.1 (-0.2 to 0.0)
Hospital Characteristics			
Hospital type, No. (%)			
Government (n = 9)	4097 (40.7)	3036 (26.8)	-13.9 (-15.1 to -12.6)
Nonprofit or charity (n = 12)	2785 (27.7)	2964 (26.2)	-1.5 (-2.7 to -0.3)
Private (n = 42)	3184 (31.6)	5308 (46.9)	15.3 (14.0 to 16.6)
Hospital size by anticipated enrollment, No. (%)			
Extra large (>1000) (n = 5)	1853 (18.4)	1707 (15.1)	-3.3 (-4.3 to -2.3)
Large (501-1000) (n = 15)	3561 (35.4)	4962 (43.9)	8.5 (7.2 to 9.8)
Medium (201-500) (n = 24)	3847 (38.2)	3568 (31.6)	-6.7 (-7.9 to -5.4)
Small (≤200) (n = 19)	805 (8.0)	1071 (9.5)	1.5 (0.7 to 2.2)
Catheterization laboratory, No. (%)			
Installed during study (n = 3)	171 (1.7)	325 (2.9)	1.2 (0.8 to 1.6)
No (n = 17)	1998 (19.8)	1554 (13.7)	-6.1 (-7.1 to -5.1)
Yes (n = 43)	7897 (78.5)	9429 (83.4)	4.9 (3.9 to 6.0)

Abbreviation: IQR, interquartile range.

SI conversions: To convert low-density lipoprotein cholesterol to mmol/L, multiply by 0.0259; triglycerides to mmol/L, multiply by 0.0113; serum creatinine to μmol/L, multiply by 88.4; and glucose to mmol/L, multiply by 0.0555.

^a Total numbers are shown for variables for which data were not completely reported.

^b Crude difference = intervention minus control.

monitoring. After study completion, 1 site was identified with an improbably high rate of reinfarction, which prompted quality assurance sampling of 10% of cases with corresponding controls among the study sites through blinded site surveys. Two additional sites with systematic error were identified in outcome reporting, which resulted in 13 changes to the primary outcome (1% of overall events).

For statistical analyses, Stata, version 14 (Stata Corp), SAS, version 9.4 (SAS Institute Inc), and R, version 3.3.0 (R Foundation), were used. StatTag, version 3.0,²³ was used for the preparation of results in this article. StatTag facilitates reproducible research by embedding output from statistical programs (Stata, SAS, and R) in Microsoft Word documents. Code files used for generation of results are available at <https://github.com/abigailbaldridge/ACS-QUIK>.

Results

Figure 1 shows the flow of hospitals and participants. We recruited 22 557 participants from 63 hospitals, including 67% private, 14% government affiliated with medical colleges, and 19% nonprofit/charity. Participants were ineligible and excluded if they had cardiac biomarker measurements that were missing or not elevated (for NSTEMI only because some patients with ST elevations and chest pain did not undergo biomarker testing [n = 954]), were transferred after thrombolytic therapy for STEMI without any further therapy (n = 132), were admitted outside the study period (n = 91), or had implausible time discrepancies (n = 6). Follow-up through 30 days was 99%. Among 21 374 eligible

Table 2. Characteristics and Association of Process of Care and Time-Based Outcomes by Treatment Group, Adjusted for Within-Hospital Clustering and Temporal Trends

	No./Total No. (%) ^a			
Measures of Care	Control (n = 10 066)	Intervention (n = 11 308)	Difference, % (95% CI) ^b	Odds Ratio or β-Coefficient (95% CI) ^b
Medications				
Prehospital aspirin	1696/10 052 (16.9)	2100/11 300 (18.6)	3.80 (2.08 to 5.52)	1.40 (1.21-1.62)
Prehospital thrombolysis (STEMI)	25/6921 (0.4)	37/6768 (0.5)	-0.04 (-0.76 to 0.68)	0.95 (0.40-2.28)
In-hospital aspirin	9858/10 042 (98.2)	11 027/11 286 (97.7)	-0.05 (-0.95 to 0.84)	0.98 (0.69-1.39)
In-hospital second antiplatelet	9861/10 050 (98.1)	11 112/11 297 (98.4)	-0.05 (-0.89 to 0.79)	0.98 (0.67-1.43)
In-hospital β-blocker	3676/9874 (37.2)	4638/10 885 (42.6)	6.25 (4.10 to 8.40)	1.46 (1.29-1.65)
In-hospital anticoagulant	8602/10 051 (85.6)	9654/11 281 (85.6)	2.60 (0.87 to 4.33)	1.27 (1.09-1.49)
Studies and procedures				
Echocardiography	9209/10 066 (91.5)	10 516/11 308 (93.0)	5.62 (3.35 to 7.89)	2.50 (1.95-3.21)
Diagnostic angiography	6181/10 066 (61.4)	6500/11 308 (57.5)	Nonestimable	Nonestimable
PCI	5282/10 066 (52.5)	5271/11 308 (46.6)	-1.66 (-3.27 to -0.04)	0.87 (0.77-0.99)
Primary PCI (STEMI)	3476/6921 (50.2)	3234/6768 (47.8)	-1.87 (-3.85 to 0.11)	0.86 (0.73-1.01)
Door-to-balloon time (STEMI), median (IQR), min	65 (53-105) (n=4022)	77 (55-118) (n=3639)	6.18 (-4.78 to 17.14)	13.00 (3.64-22.36) ^c
Thrombolysis (STEMI)	1596/6921 (23.1)	1571/6768 (23.2)	5.56 (3.21 to 7.91)	1.59 (1.33-1.92)
Door-to-needle time (STEMI), median (IQR), min	44 (30-67) (n=1455)	45 (27-75) (n=1433)	0.40 (-5.23 to 6.03)	5.00 (-5.20-15.20) ^c
Any reperfusion (STEMI)	5067/6921 (73.2)	4805/6768 (71.0)	3.43 (0.87 to 6.00)	1.24 (1.06-1.46)
Rescue PCI (STEMI)	644/6891 (9.3)	1031/6767 (15.2)	2.45 (0.51 to 4.38)	1.39 (1.08-1.79)
Discharge treatment and counseling				
Discharge aspirin	8777/8998 (97.5)	10 360/10 559 (98.1)	1.36 (0.23 to 2.49)	1.65 (1.15-2.37)
Discharge second antiplatelet ^d	8824/9011 (97.9)	10 377/10 580 (98.1)	0.41 (-0.31 to 1.13)	1.24 (0.86-1.79)
Discharge β-blocker	5808/8894 (65.3)	6799/10 178 (66.8)	6.69 (4.43 to 8.95)	1.48 (1.30-1.68)
Discharge statin	8700/9006 (96.6)	10 289/10 579 (97.3)	1.21 (0.07 to 2.35)	1.42 (1.04-1.92)
Discharge ACE inhibitor or ARB	534/1029 (51.9)	643/1495 (43.0)	6.57 (0.58 to 12.57)	1.45 (1.03-2.04)
Cardiac rehabilitation referral	2362/9228 (25.6)	3322/10 791 (30.8)	Nonestimable	Nonestimable

Abbreviations: ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; IQR, interquartile range; PCI, percutaneous coronary intervention; STEMI, ST-segment elevation myocardial infarction.

^a Data are expressed as No./total No. of participants unless otherwise indicated. Totals include eligible participants for whom process measures were reported. Patients with contraindications to medications were excluded from relevant process measures. Discharge measures were assessed only among patients discharged. Discharge ACE inhibitors or ARBs were assessed only among discharged patients with ejection fractions of 40% or lower.

^b Odds ratios and β -coefficients represent effect of intervention compared with control and are calculated as the difference in marginal effects (intervention group minus control group) in a mixed-effects logistic regression or quantile regression model including a random-effects term to account for within-hospital clustering and temporal trends.

^c β -Coefficient.

^d Second antiplatelet includes clopidogrel, prasugrel, or ticagrelor.

participants, 295 (1%) had incomplete outcome data (215 patients were not followed up by the enrolling site and 80 patients could not be reached after 3 contact attempts). Unadjusted baseline characteristics between included participants and participants missing follow-up data were generally similar except for a lower initial troponin level in the former group (1.3 ng/mL vs 4.6 ng/mL; $P < .001$) (eTable 1 in Supplement 3). The intervention and control groups included 11 308 (53%) and 10 066 (47%) participants, respectively.

Participants

Table 1 shows the baseline characteristics for the intervention and control groups. The mean age of the participants was 60.6 years, 76% were men, 31% had a history of tobacco use, and 44% had a history of diabetes mellitus. Sixty-four percent of participants presented with STEMI, with a median (symptom-to-door time of 240 (interquartile range, 120-840)

minutes in the control group compared with 255 (interquartile range, 111-825) minutes in the intervention group.

Table 2 shows the association of process of care and time-based characteristics with the intervention, both unadjusted and adjusted for within-hospital clustering and temporal trends. Compared with the control group, the intervention group had a higher rate of prehospital aspirin use (18.6% vs 16.9%; adjusted risk difference, 3.80% [95% CI, 2.08%-5.52%]; adjusted odds ratio [OR], 1.40 [95% CI, 1.21-1.62]) but similarly low rates of prehospital thrombolysis (<1%). Among eligible individuals without contraindications, rates of in-hospital aspirin use and a second antiplatelet medication (clopidogrel, prasugrel, or ticagrelor) were high (>95%) and similar in both groups. Compared with the control group, the intervention group had a higher rate of in-hospital β -blocker use (43% vs 37%; adjusted risk difference, 6.25% [95% CI, 4.10%-8.10%]; adjusted OR, 1.46 [95%

Table 3. Unadjusted and Adjusted Primary and Secondary Trial Outcomes Using Mixed-Effects Logistic Regression Models That Account for Within-Hospital Clustering and Clustering and Temporal Trends

Outcomes	No. (%) Control (n = 10 066)	Intervention (n = 11 308)	Cluster-Adjusted Difference, % (95% CI) ^a	Cluster-Adjusted Odds Ratio (95% CI) ^a	Primary Analysis Difference, % (95% CI) ^a	Primary Analysis Odds Ratio (95% CI) ^a	ICC
Primary outcome							
30-d MACE	645 (6.4)	602 (5.3)	-0.51 (-1.28 to 0.26)	0.92 (0.81-1.04)	-0.09 (-1.32 to 1.14)	0.98 (0.80-1.21)	0.15
Secondary outcomes							
30-d mortality	509 (5.1)	445 (3.9)	-0.65 (-1.34 to 0.03)	0.87 (0.75-1.00)	-0.28 (-1.35 to 0.80)	0.94 (0.74-1.19)	0.18
30-d cardiovascular mortality	494 (4.9)	434 (3.8)	-0.58 (-1.24 to 0.09)	0.88 (0.76-1.02)	-0.26 (-1.31 to 0.80)	0.94 (0.74-1.20)	0.19
In-hospital mortality	331 (3.3)	321 (2.8)	-0.05 (-0.58 to 0.47)	0.98 (0.82-1.17)	-0.23 (-1.07 to 0.60)	0.93 (0.70-1.22)	0.18
30-d reinfarction	121 (1.2)	135 (1.2)	0.12 (-0.31 to 0.55)	1.08 (0.82-1.42)	0.50 (-0.24 to 1.24)	1.39 (0.87-2.22)	0.33
30-d stroke	60 (0.6)	90 (0.8)	0.20 (-0.05 to 0.45)	1.34 (0.94-1.93)	0.14 (-0.23 to 0.52)	1.24 (0.71-2.15)	0.14
30-d major GUSTO bleeding ^b	19 (0.2)	30 (0.3)	0.13 (-0.05 to 0.30)	1.56 (0.84-2.88)	0.23 (-0.05 to 0.52)	2.34 (0.93-5.89)	0.20
Optimal in-hospital medication ^c	3122 (31.7)	3878 (35.8)	8.61 (6.98 to 10.23)	1.70 (1.57-1.85)	6.00 (3.90 to 8.11)	1.45 (1.28-1.64)	0.40
Optimal discharge medication ^d	5454 (61.8)	6483 (64.0)	9.97 (8.32 to 11.61)	1.73 (1.59-1.87)	8.66 (6.30 to 11.03)	1.61 (1.42-1.82)	0.28
Tobacco cessation advice ^e	3526 (96.0)	2618 (94.7)	0.30 (-1.20 to 1.80)	1.06 (0.80-1.39)	0.30 (-2.15 to 2.76)	1.06 (0.67-1.67)	0.37

Abbreviations: ICC, intraclass correlation; MACE, major adverse cardiovascular events, defined as death, reinfarction, stroke, and major GUSTO bleeding.

^a Odds ratios represent effect of intervention compared with control and are calculated as the difference in marginal effects (intervention group minus control group) in a mixed-effects logistic regression model including a random-effects term to account for within-hospital clustering. Primary analysis additionally accounted for temporal trends.

^b Major bleeding is defined by the Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries (GUSTO) criteria,

which is defined by intracerebral hemorrhage or bleeding resulting in substantial hemodynamic compromise requiring treatment.

^c Composed of aspirin, adenosine diphosphate receptor antagonist (clopidogrel, prasugrel, or ticagrelor), anticoagulant, and β -blocker among patients eligible to receive all medications.

^d Composed of aspirin, adenosine diphosphate receptor antagonist (clopidogrel, prasugrel, or ticagrelor), statin, and β -blocker among discharged patients eligible to receive all medications.

^e Among discharged patients who reported smoking at baseline.

CI, 1.29-1.65)) and anticoagulant use (86% vs 86%; adjusted risk difference, 2.60% [95% CI, 0.87%-4.33%]; adjusted OR, 1.27 [95% CI, 1.09-1.49]).

Compared with the control group, the intervention group had a higher rate of echocardiography (93% vs 92%; adjusted risk difference, 5.62% [95% CI, 3.35%-7.89%]; adjusted OR, 2.50 [95% CI, 1.95-3.21]), and among patients with STEMI, higher rates of thrombolysis (23% vs 23%; adjusted risk difference, 5.56% [95% CI, 3.21%-7.91%]; adjusted OR, 1.59 [95% CI, 1.33-1.92]), reperfusion (71% vs 73%; adjusted risk difference, 3.43% [95% CI, 0.87%-6.00%]; adjusted OR, 1.24 [95% CI, 1.06-1.46]), and rescue percutaneous coronary intervention (15% vs 9%; adjusted risk difference, 2.45% [95% CI, 0.51%-4.38%]; adjusted OR, 1.39 [95% CI, 1.08-1.79]) but lower rates of diagnostic coronary angiography (58% vs 61%) and percutaneous coronary intervention (47% vs 53%; adjusted risk difference, -1.66% [95% CI, -3.27% to -0.04%]; adjusted OR, 0.87 [95% CI, 0.77-0.99]).

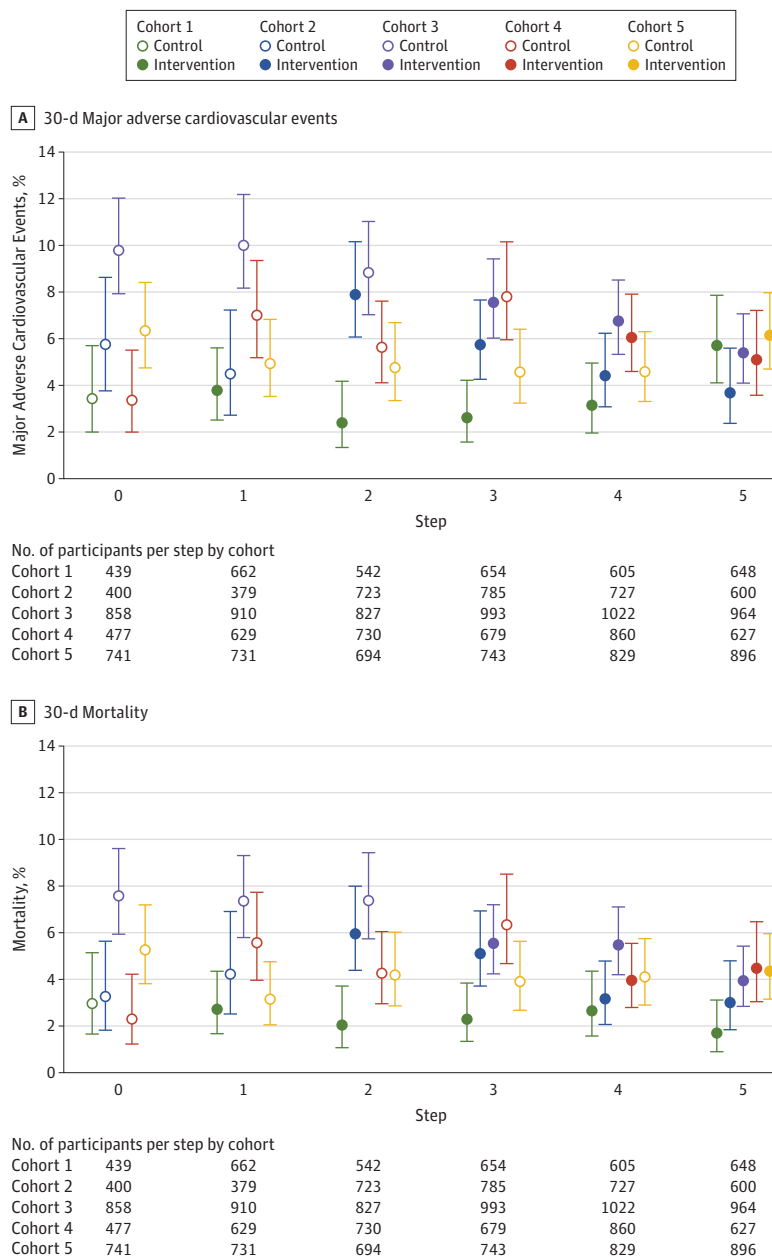
The intervention group had a higher rate of discharge aspirin use (98% vs 98%; adjusted risk difference, 1.36% [95% CI, 0.23%-2.49%]; adjusted OR, 1.65 [95% CI, 1.15-2.37]), β -blocker use (67% vs 65%; adjusted risk difference, 6.63% [95% CI, 4.43%-8.95%]; adjusted OR, 1.47 [95% CI, 1.30-1.68]), statin use (97% vs 97%; adjusted risk difference, 1.21% [95% CI, 0.07%-2.35%]; adjusted OR, 1.42 [95% CI, 1.04-1.92]), angiotensin-converting enzyme inhibitor or angiotensin receptor blocker use among patients with ejection frac-

tions of 40% or lower (43% vs 52%; adjusted risk difference, 6.57% [95% CI, 0.58%-12.57%]; adjusted OR, 1.45 [95% CI, 1.03-2.04]), and referral for cardiac rehabilitation (31% vs 26%). Rates of discharge second antiplatelet use and tobacco cessation counseling were high (>95%) and similar between groups.

Primary Analysis

Table 3 shows the cluster-adjusted and cluster- and temporal-adjusted primary and secondary trial outcome event rates using mixed-effects logistic regression models that account for within-hospital clustering and temporal trends. The unadjusted rate of 30-day major adverse cardiovascular events was 5.3% in the intervention group compared with 6.4% in the control group. **Figure 2A** shows the temporal trends for 30-day major adverse cardiovascular events by group and step. The cluster-adjusted OR for 30-day major adverse cardiovascular events was 0.92 (95% CI, 0.81-1.04), which was not statistically significant after further adjustment for temporal trends (adjusted risk difference, -0.09% [95% CI, -1.32% to 1.14%]; adjusted OR, 0.98 [95% CI, 0.80-1.21]). Results were similar after multiple imputation of data from individuals who had missing follow-up data or were lost to follow-up. After sensitivity analyses of hypothetical scenarios to account for the potential effects of missing follow-up events and using multiple imputation, the range of results varied more broadly, but overall plausible results were consistent with the reported primary analysis.

Figure 2. Unadjusted Temporal Trends in 30-Day Major Adverse Cardiovascular Event Rates and Mortality Rates



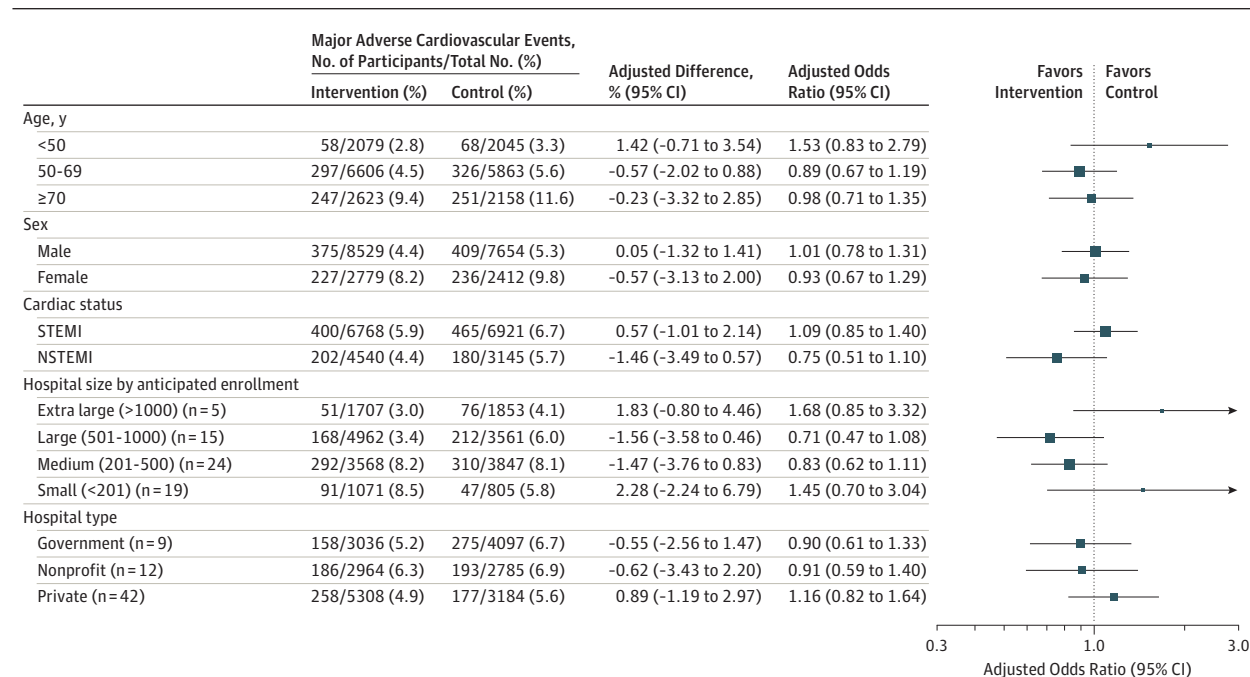
Open circles indicate rates during the control period; solid circles represent rates during the intervention period. Error bars indicate 95% CIs.

Secondary and Post Hoc Analyses

The rate of 30-day death was 3.9% in the intervention group compared with 5.1% in the control group. Figure 2B shows the temporal trends for 30-day death by group and step. The cluster-adjusted OR for 30-day death was 0.87 (95% CI, 0.75-1.00), an effect that was not statistically significant after adjusting for temporal trends (adjusted risk difference, -0.28% [95% CI, -1.35% to 0.80%]; adjusted OR, 0.94 [95% CI, 0.74-1.19]). These results did not materially change after adjustment for GRACE score covariates (eTable 5 in Supplement 3). In-hospital outcomes demonstrated a similar pattern overall. We also tested post hoc for a potential interaction for length

of time exposed to the intervention to evaluate whether the intervention might have been more effective if given over a longer period, and the overall effects were similar (eTable 6 in Supplement 3).

In a post hoc analysis, we evaluated the effect of the intervention on the expanded outcome of 30-day major adverse cardiovascular event plus incident in-hospital heart failure, cardiogenic shock, or cardiac arrest (eTable 7 in Supplement 3). The unadjusted rate was 7.0% in the intervention group and 9.1% in the control group. The cluster-adjusted OR for this outcome was 0.89 (95% CI, 0.80-1.00), which was similar after temporal adjustment (adjusted risk

Figure 3. Effect of Quality Improvement Tool Kit Intervention on 30-Day Major Adverse Cardiovascular Events by Prespecified Subgroups After Adjustment for Within-Hospital Clustering and Temporal Trends

NSTEMI indicates non-ST-segment elevation myocardial infarction; STEMI, ST-segment elevation myocardial infarction. Size of the data markers indicates precision of the point estimates.

difference, -1.34% [95% CI, -2.72% to 0.04%]; adjusted OR, 0.84 [95% CI, 0.70-1.00]).

Figure 3 shows the effect of the intervention on the rate of 30-day major adverse cardiovascular events by prespecified subgroups, including age, sex, STEMI vs NSTEMI, hospital size, and hospital type. We found a similar pattern of effect across subgroups. eFigure 1A in [Supplement 3](#) shows the heterogeneity of the primary outcome by hospital through a residual plot, which evaluates the event rate by site compared with the mean event rate throughout the entire trial. eFigure 1B in [Supplement 3](#) shows the unadjusted within-hospital difference in the rate of 30-day composite major adverse cardiovascular events between the intervention and control groups. eFigures 2A and 2B in [Supplement 3](#) show temporal changes in the primary outcome and 30-day mortality rates within and among hospitals.

Discussion

The use of a locally adapted quality improvement tool kit did not reduce the primary outcome of death, reinfarction, stroke, or major bleeding at 30 days among patients presenting with acute myocardial infarction to Kerala hospitals. Major adverse cardiovascular event rates were lower than previously estimated from Kerala, and in-hospital and discharge treatment rates were higher, which may have influenced the results.⁴ Symptom-to-door times were also shorter than in previous reports from India.^{4,24} Hospitals' previous participation in the Kerala ACS Registry⁴ or contamination

between intervention and control groups may have also influenced these results through improved care throughout the state over the study period. However, the intervention may not have been effective because of insufficient training, implementation or adoption of the intervention, or period of exposure to sufficiently change hospital practice, including among process measures such as speed of reperfusion therapy. Although the absolute event rate was lower in the intervention group, the stepped-wedge trial design demonstrates the importance of accounting for temporal trends for this type of intervention.

The intervention led to lower 30-day major adverse cardiovascular event and mortality rates in the first step compared with other steps. Potential reasons for this observation include (1) smaller sample size and instability of the estimate; (2) lack of contamination; (3) baseline participant- or hospital-level characteristics; or (4) chance. The intervention led to a higher rate of in-hospital use of reperfusion, thrombolytic therapy, and anticoagulation and a higher rate of discharge aspirin, β -blocker, statin, and angiotensin-converting enzyme inhibitor or angiotensin receptor blocker prescriptions. A post hoc analysis demonstrated that the intervention led to a lower rate of expanded major adverse cardiovascular events, which also included in-hospital heart failure, cardiogenic shock, and cardiac arrest, but other secondary clinical outcomes were not different between groups. It is possible that these post hoc outcome findings were driven by the improvements in medication use, including reperfusion, but they may also be due to recruitment or detection bias or chance and should be considered only hypothesis generating.

The primary outcome results were consistent across pre-specified individual- and hospital-level subgroups despite heterogeneity of effect among hospitals. This heterogeneity may have been driven by site-level personnel, intervention fidelity, recruitment or detection bias, or chance but requires future research.

Previous cluster randomized trials of acute coronary syndrome quality improvement in low- and middle-income countries, including the Brazilian Intervention to Increase Evidence Usage in Acute Coronary Syndromes (BRIDGE-ACS),⁹ and Clinical Pathways for Acute Coronary Syndrome-Phase 2 (CPACS-2),²⁵ demonstrated improvements in composite medication use and reperfusion among patients with STEMI. These smaller trials did not demonstrate statistically significant improvements in clinical outcomes but were also not powered to detect such differences. However, these trials also demonstrated lower intraclass correlation coefficients than in the present study and did not use a stepped-wedge design to account for temporal trends. The recently completed Clinical Pathways for Acute Coronary Syndrome-Phase 3 (CPACS-3) trial enrolled more than 29 000 Chinese participants using a cluster randomized, stepped-wedge design may be helpful for understanding the effects of quality improvement interventions in low- and middle-income country settings.²⁵ Results from the present trial contrast with the favorable temporal trends in clinical outcomes that have been demonstrated in nonrandomized quality improvement and health system intervention studies, including a 2017 nonrandomized trial¹¹ designed to increase primary percutaneous coronary intervention use in Tamil Nadu. These differences may be related to study design, intervention targets, components, training and implementation, comparator group outcomes, or a combination thereof. Event rates in this trial were similar to outcomes reported from the American College of Cardiology/American Heart Association Acute Coronary Treatment and Intervention Outcomes Network-Get With the Guidelines (ACTION-GWTG) program in 2014 (in-hospital mortality, 6.4% for STEMI and 3.4% for NSTEMI).²⁶ However, there are important gaps in guideline-directed care that remain in acute myocardial infarction care in Kerala, including rate and speed of reperfusion, and likely gaps in other acute cardiovascular conditions (eg, stroke, heart failure), other states in India, and other low- and middle-income countries that warrant further study.

This study has several strengths. First, this trial built on previous observational data from the Kerala Acute Coronary Syndrome Registry and collaborated with the Cardiological Society of India-Kerala Chapter to execute the largest randomized cardiovascular intervention trial in India to date. Second, this study used a trial design with appropriate statistical methods to improve internal validity of the results. Third, this trial used advanced yet low-cost trial monitoring procedures, including central statistical, risk-based monitoring, to capture high-quality data from many sites that had not previously participated in clinical trials.

Limitations

This study has several limitations. First, the complex intervention included several evidence-based components (eg, audit and feedback to improve process measures,⁶ checklists to reduce errors⁸) that may not have been fully implemented at all sites or for a sufficient duration to improve clinical outcomes. Understanding the external and internal conditions that were associated with improved intervention implementation and outcomes requires further research. Second, the trial was susceptible to recruitment bias because, while randomization occurred at the cluster level, informed consent was required from individual participants for 30-day follow-up. However, baseline differences in key covariates associated with the primary outcome were limited. Third, these results demonstrated higher-than-anticipated intraclass correlation coefficients than previous trials and pretrial estimates, which reduced statistical power. However, it seems unlikely that this would have materially changed the primary outcome estimate. Fourth, quality of clinical care was higher and event rates were lower than anticipated in the trial. It is uncertain whether such an intervention would be effective in an environment where the quality of care is lower or the event rates are higher. Fifth, secondary analyses were not adjusted for multiple comparisons, which increases the possibility of type I error.

Conclusions

Among patients with acute myocardial infarction in Kerala, India, use of a quality improvement intervention compared with usual care did not decrease a composite of 30-day major adverse cardiovascular events. Further research is needed to understand the lack of efficacy.

ARTICLE INFORMATION

Accepted for Publication: January 11, 2018.

Author Affiliations: Northwestern University Feinberg School of Medicine, Chicago, Illinois (Huffman, Baldrige, Lloyd-Jones); WestFort Hi-Tech Hospital Ltd, Thrissur, India (Mohan); Centre for Chronic Disease Control, Gurgaon, India (Devarajan, Kondal, Zhao, Ali, Prabhakaran); Public Health Foundation of India, Gurgaon, India (Devarajan, Kondal, Zhao, Ali, Prabhakaran); Medical College Hospital, Kozhikode, India (Krishnan); District Hospital, Kollam, India (Natesan); Amala Medical College,

Thrissur, India (Gopinath); Medical College Hospital, Alappuzha, India (Viswanathan); Little Flower Hospital, Angamaly, India (Stigi); Caritas Hospitals, Kottayam, India (Joseph); EMS Hospital, Perinthalmanna, India (Chozhakkat); London School of Hygiene and Tropical Medicine, London, England (Prabhakaran).

Author Contributions: Drs Huffman and Kondal had full access to all study data and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Huffman and Mohanan are co-first authors.

Concept and design: Huffman, Devarajan, Kondal, Krishnan, Lloyd-Jones, Prabhakaran.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Huffman, Kondal, Ali, Lloyd-Jones.

Critical revision of the manuscript for important intellectual content: Huffman, Mohanan, Devarajan, Baldrige, Zhao, Krishnan, Natesan, Gopinath, Viswanathan, Stigi, Joseph, Chozhakkat, Lloyd-Jones, Prabhakaran.

Statistical analysis: Huffman, Baldrige, Kondal, Zhao.

Obtained funding: Huffman, Mohanan, Lloyd-Jones, Prabhakaran.

Administrative, technical, or material support: Huffman, Mohanan, Devarajan, Baldrige, Lloyd-Jones, Prabhakaran.

Supervision: Huffman, Mohanan, Lloyd-Jones, Prabhakaran.

Conflict of Interest Disclosures: All authors have completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Dr Huffman reports receipt of support from the World Heart Federation to serve as its senior program advisor for the Emerging Leaders program, which is supported by unrestricted educational grants from Boehringer Ingelheim and Novartis with previous support from BUPA and AstraZeneca.

Group Information: *Cardiological Society of India Kerala Chapter Advisory Committee:* G. Vijayaraghavan, Kerala Institute of Medical Sciences; Ashokan Nambiar, Baby Memorial Hospital; Geevar Zachariah, Mother Hospital; Rajan J. Manjooran, Pushpagiri Medical College; S. Sivasankaran, Sree Chitra Tirunal Institute of Medical Science and Technology. *Trial steering committee:* Dorairaj Prabhakaran (cochair), Centre for Chronic Disease Control/Public Health Foundation of India; Donald M. Lloyd-Jones (cochair), Northwestern University; K. Srinath Reddy, Public Health Foundation of India; S. Harikrishnan, Sree Chitra Tirunal Institute of Medical Science and Technology; Robert O. Bonow, Northwestern University; Darwin R. Labarthe, Northwestern University; Sidney C. Smith Jr, University of North Carolina. *Data and safety monitoring board:* Brahmajee Nallamothu (chair), University of Michigan; Thomas Alexander, Kovai Medical Center and Hospital; Karla Hemming, University of Birmingham; Simon Thom, Imperial College London; K. R. Sundaram, Amrita Institute of Medical Sciences; Lawton Cooper, National Heart, Lung, and Blood Institute. *Operations committee:* Mark D. Huffman, Northwestern University; P. P. Mohanan, Westfort Hi-Tech Hospital Ltd; Raji Devarajan, Centre for Chronic Disease Control; Abigail S. Baldrige, Northwestern University; Dimple Kondal, Centre for Chronic Disease Control; Lihui Zhao, Northwestern University; Mumtaj Ali, Public Health Foundation of India. *Zonal project coordinators:* Divin Davies, WestFort Hi-Tech Hospital; Prasad Arumugan, Caritas Hospital; Theckethottathil Chacko Aneesh, PRS Hospital. *Trial sites, investigators, and coordinators:* Amala Medical College, Thrissur: Dr Rajesh, Mr Anoop; Amrita Institute of Medical Sciences & Research Center, Ernakulam: Dr Natarajan, Mr Sujith Raj; Anathapurai Hospital, Thiruvananthapuram: Dr Bahuleyan, Mr Jinbert; Aswini Hospital Thrissur: Dr Pramod, Mr Vineesh Varghese; Baby Memorial Hospital, Kozhikode: Dr Nambiar, Dr Bindu; Bharath Heart Institute, Kottayam: Dr Kathalankal, Mrs Susamma; Bishop Benziger Hospital, Kollam: Dr Renga, Ms Alphonsa; CH Memorial Hospital, Valanchery: Dr Ibrahimkutty; Caritas Hospitals: Dr Joseph, Mr Tony; Daya Specialty Hospital, Thrissur: Dr Ullas, Mr Ajmal; District Hospital, Palakkad: Dr Siar, Ms Pravya; District Hospital, Kollam: Dr Syam, Mrs Sajitha; Dr Damodaran Memorial Hospital, Kollam: Dr Robby, Mrs Prasanna; Elite Mission Hospital, Thrissur: Dr Manikandan, Ms Lekha; EMS Hospital Perinthalmanna: Dr Somanathan; Gokulam Medical College, Thiruvananthapuram: Dr Abilash, Dr Binu,

Mr Aneesh; Government Medical College Thrissur: Dr Mathew, Dr Andrews, Mr Arun Gopi; Holy Cross Hospital, Kottayam: Dr Chacko, Mr Libin; Indira Gandhi Cooperative Hospital, Ernakulam: Dr Abraham, Ms Alphonsa Rony; Irinjalakuda Co-operative Hospital: Dr Ullas, Mr Midhun George; Jubilee Mission Hospital, Thrissur: Dr Govindanunni, Mr Lance Frank William; Kannur Medical College, Kannur: Dr Raveendran, Ms Athira; Kerala Institute of Medical Science, Thiruvananthapuram: Dr Vijayaraghavan, Mrs Kavitha; KMCT Heart Institute, Manassery: Dr Saleem, Mr Joshy; Koyili Hospital, Kannur: Dr Kumar, Ms Alpha; KVM Hospital, Cherthala: Dr Venugopal, Mr Vipin; Lakshmi Hospital, Palakkad: Dr Jayagopal, Ms Devaki; Lakshmi Hospital, Ernakulam: Dr Sasikumar; Lal Memorial Hospital, Irinjalakuda: Dr Madhu; Lisie Hospital, Ernakulam: Dr Abdullakutty, Dr Mathew, Ms Serrin; Little Flower Hospital, Angamaly: Dr Joseph, Mr Rajesh; Lourdes Hospital, Ernakulam: Dr Sujith Kumar, Mrs Ria Sandeep; Medical College Hospital, Kozhikode: Dr Haridas, Mrs Deepa; Medical College Hospital, Alappuzha: Dr Viswanathan, Dr Sivaprasad, Dr Sreenivas, Dr Gagan; Medical College Thiruvananthapuram: Dr Koshy, Dr Raji; Medical College, Kottayam: Dr Jayaprakash, Dr Brijesh, Dr Brjesh; Metro International Cardiac Centre, Kozhikode: Dr Mustafa, Ms Anooja; MIMS Heart, Kottakkal, Malappuram: Dr Tahsin, Mr Pradeesh; Modern Hospital, Kodungallur: Dr Ukken, Ms Teena Sudheer; MOSCM Hospital, Ernakulam: Dr Punnoose, Mr Binoy Kurian; Mother Hospital Limited, Thrissur: Dr K. J. James; NIMS Hospital, Thiruvananthapuram: Dr Sreedharan, Mrs Anju Mohan; Pariyaram Medical College, Kannur: Dr Sebastian, Mr Robin; PRS Hospital, Thiruvananthapuram: Dr Nair, Mr Aneesh; Pushpagiri Medical College, Thiruvilla, Dr Manjooran, Mr Jacob; PVS Memorial Hospital, Ernakulam: Dr Blessan, Mrs Nisha; Rajah Hospital: Guruvayoor, Dr Showjad, Dr Dharmoadharan; Ramdas Nursing Home, Perinthalmanna: Dr Ramadas, Mr Jobson; SK Hospital, Thiruvananthapuram: Dr Suresh, Mrs Divy; SH Medical Centre Hospital, Kottayam: Dr Chacko, Mrs Saranya; Samaritan Hospital, Pazhangad: Dr Eapen, Mrs Sindhu; Santhi Nursing Home, Punnayoorakulam: Dr Shaji, Dr Krishnan; Sree Chitra Tirunal Institute of Medical Sciences & Technology, Thiruvananthapuram: Dr Harikrishnan, Dr Ajit, Mr Suresh; Sree Narayana Institute of Medical Science, Ernakulam: Dr Menon, Mrs Nisha; St James Hospital, Chalakudy: Dr Jubil Mathew; St Joseph Hospital, Dharmagiri: Dr Thomas, Sr Betto; St Martin De Porres Hospital, Cherukunnu: Dr Muralidharan; St Mary's Hospital, Thodupuzha: Dr Mathew Abraham; Sukuppuram Hospital, Edappal: Dr Narayanan; Tellicherry Co-operative Hospital, Thalassery: Dr Kumar, Dr Manoj; Thangam Hospital, Palakkad: Dr Jayakumar, Mr Sagar Thampy; Travancore Medical College, Thiruvananthapuram: Dr Abhilash, Mr Santhosh; WestFort Hi-Tech Hospital Limited, Thrissur: Dr E. B. Manoj, Mr Divin Davies.

Funding/Support: This project was funded by the National Heart, Lung, and Blood Institute (grant R00 HL107749) with support from the Cardiological Society of India-Kerala Chapter, Centre for Chronic Disease Control, Northwestern University Global Health Initiative, and

Northwestern University Clinical and Translational Science Institute (grant ULTR001422).

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; or decision to submit the manuscript for publication.

REFERENCES

1. Roth GA, Johnson C, Abajobir A, et al. Global, regional, and national burden of cardiovascular diseases for 10 causes, 1990 to 2015. *J Am Coll Cardiol*. 2017;70(1):1-25.
2. Cohen MG, Fonarow GC, Peterson ED, et al. Racial and ethnic differences in the treatment of acute myocardial infarction: findings from the Get With the Guidelines-Coronary Artery Disease program. *Circulation*. 2010;121(21):2294-2301.
3. Buckley GJ, Pittluck RE; Board on Global Health, Institute of Medicine, National Academies of Sciences, Engineering, and Medicine. *Improving Quality of Care in Low- and Middle-Income Countries*. Washington, DC: National Academies Press; 2015.
4. Mohanan PP, Mathew R, Harikrishnan S, et al; Kerala ACS Registry Investigators. Presentation, management, and outcomes of 25 748 acute coronary syndrome admissions in Kerala, India: results from the Kerala ACS Registry. *Eur Heart J*. 2013;34(2):121-129.
5. Huffman MD, Prabhakaran D, Abraham AK, Krishnan MN, Nambiar AC, Mohanan PP; Kerala Acute Coronary Syndrome Registry Investigators. Optimal in-hospital and discharge medical therapy in acute coronary syndromes in Kerala: results from the Kerala Acute Coronary Syndrome Registry. *Circ Cardiovasc Qual Outcomes*. 2013;6(4):436-443.
6. Ivers N, Jamtvedt G, Flottorp S, et al. Audit and feedback: effects on professional practice and healthcare outcomes. *Cochrane Database Syst Rev*. 2012;(6):CD000259.
7. Rotter T, Kinsman L, James E, et al. Clinical pathways: effects on professional practice, patient outcomes, length of stay and hospital costs. *Cochrane Database Syst Rev*. 2010;(3):CD006632.
8. Ko HC, Turner TJ, Finnigan MA. Systematic review of safety checklists for use by medical care teams in acute hospital settings—limited evidence of effectiveness. *BMC Health Serv Res*. 2011;11(1):211.
9. Berwanger O, Guimarães HP, Laranjeira LN, et al; BRIDGE-ACS. A multifaceted intervention to narrow the evidence-based gap in the treatment of acute coronary syndromes: rationale and design of the Brazilian Intervention to Increase Evidence Usage in Acute Coronary Syndromes (BRIDGE-ACS) cluster-randomized trial. *Am Heart J*. 2012;163(3):323-329.
10. Du X, Gao R, Turnbull F, et al; CPACS Investigators. Hospital quality improvement initiative for patients with acute coronary syndromes in China: a cluster randomized, controlled trial. *Circ Cardiovasc Qual Outcomes*. 2014;7(2):217-226.
11. Alexander T, Mulasari AS, Joseph G, et al. A system of care for patients with ST-segment elevation myocardial infarction in India: the Tamil Nadu-ST-Segment Elevation Myocardial Infarction Program. *JAMA Cardiol*. 2017;2(5):498-505.

12. Huffman MD, Mohanan PP, Devarajan R, et al. Acute coronary syndrome quality improvement in Kerala (ACS QUIK): rationale and design for a cluster-randomized stepped-wedge trial. *Am Heart J*. 2017;185:154-160.
13. Thygesen K, Alpert JS, Jaffe AS, et al; Joint ESC/ACCF/AHA/WHF Task Force for the Universal Definition of Myocardial Infarction. Third universal definition of myocardial infarction. *Circulation*. 2012;126(16):2020-2035.
14. Cannon CP, Brindis RG, Chaitman BR, et al; American College of Cardiology Foundation/American Heart Association Task Force on Clinical Data Standards; American College of Emergency Physicians; Emergency Nurses Association; National Association of Emergency Medical Technicians; National Association of EMS Physicians; Preventive Cardiovascular Nurses Association; Society for Cardiovascular Angiography and Interventions; Society of Cardiovascular Patient Care; Society of Thoracic Surgeons. 2013 ACCF/AHA key data elements and definitions for measuring the clinical management and outcomes of patients with acute coronary syndromes and coronary artery disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Clinical Data Standards (Writing Committee to Develop Acute Coronary Syndromes and Coronary Artery Disease Clinical Data Standards). *Circulation*. 2013;127(9):1052-1089.
15. Moore GF, Audrey S, Barker M, et al. Process evaluation of complex interventions: Medical Research Council guidance. *BMJ*. 2015;350:h1258.
16. GUSTO Investigators. An international randomized trial comparing four thrombolytic strategies for acute myocardial infarction. *N Engl J Med*. 1993;329(10):673-682.
17. Wood L, Egger M, Gluud LL, et al. Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ*. 2008;336(7644):601-605.
18. Kirkwood AA, Cox T, Hackshaw A. Application of methods for central statistical monitoring in clinical trials. *Clin Trials*. 2013;10(5):783-806.
19. Berwanger O, Guimarães HP, Laranjeira LN, et al; Bridge-ACS Investigators. Effect of a multifaceted intervention on use of evidence-based therapies in patients with acute coronary syndromes in Brazil: the BRIDGE-ACS randomized trial. *JAMA*. 2012;307(19):2041-2049.
20. Peterson ED, Roe MT, Mulgund J, et al. Association between hospital process performance and outcomes among patients with acute coronary syndromes. *JAMA*. 2006;295(16):1912-1920.
21. Granger CB, Goldberg RJ, Dabbous O, et al; Global Registry of Acute Coronary Events Investigators. Predictors of hospital mortality in the Global Registry of Acute Coronary Events. *Arch Intern Med*. 2003;163(19):2345-2353.
22. Hemming K, Girling A. A menu-driven facility for power and detectable-difference calculations in stepped-wedge cluster-randomized trials. *Stata J*. 2014;14(2):363-380.
23. Welty LJ, Rasmussen LV, Baldrige AS, et al. *StatTag*. Chicago, IL: Galtier Health Sciences Library; 2016. doi:10.18131/G36K76
24. Xavier D, Pais P, Devereaux PJ, et al; CREATE Registry Investigators. Treatment and outcomes of acute coronary syndromes in India (CREATE): a prospective analysis of registry data. *Lancet*. 2008;371(9622):1435-1442.
25. Li S, Wu Y, Du X, et al; CPACS-3 Investigators. Rational and design of a stepped-wedge cluster randomized trial evaluating quality improvement initiative for reducing cardiovascular events among patients with acute coronary syndromes in resource-constrained hospitals in China. *Am Heart J*. 2015;169(3):349-355.
26. Masoudi FA, Ponirakis A, de Lemos JA, et al. Trends in US cardiovascular care: 2016 report from 4 ACC national cardiovascular data registries. *J Am Coll Cardiol*. 2017;69(11):1427-1450.